

CHARLTON SMITH

M.S. Software Engineering

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SUMMARY

Software engineer with 7 years of experience in enterprise and startup environments. Currently building AI-powered applications and tools. Background includes co-founding a fitness startup and running an automated NFT trading operation. I like building things that work.

azoni.ai/chat — Chatbot that answers questions about my background. Paste a job description for fit analysis.

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, SQL, C#

AI/ML: OpenAI APIs, Anthropic Claude API, RAG (embeddings + vector search), Prompt Engineering

Frontend: React, Vite, HTML, CSS, Responsive Design

Backend: Node.js, FastAPI, Django, REST APIs

Cloud: AWS (Lambda, EC2, S3, CloudWatch), Docker, CI/CD, Netlify, Render

Databases: PostgreSQL, MongoDB, Redis, Firebase, SQLite

PROFESSIONAL EXPERIENCE

Independent Software Engineer | Self-Employed — *Seattle, WA*

Nov 2023 – Present

- Building AI-powered applications and developer tools — see Personal Projects section for details

Senior Software Engineer | Capital One — *Remote*

Nov 2022 – Nov 2023

- Maintained and improved automated testing pipeline for customer email notifications — test cases stored as JSON in S3, executed via AWS Lambda, results logged to CloudWatch
- Designed JSON schema for test cases so new tests could be added without changing the pipeline code
- Added Slack and email integrations for failure alerts so engineers knew about issues faster
- Mentored summer intern through project scoping, code reviews, and deployment; intern shipped an internal automation tool

Founder | Dustbunny — *Seattle, WA*

Sep 2021 – Jun 2022

- Built and operated automated NFT bidding system distributed across 50 machines, processing about 2,500 bids per minute
- Polled floor prices into Redis for sub-second lookups. Bidding logic included max bid safeguards and collection liquidity analysis
- Integrated OpenSea SDK for order execution and Etherscan API for blockchain data. Deployed custom ERC-721 contract via Remix

Software Engineer II | T-Mobile — *Bellevue, WA*

Jun 2018 – Apr 2022

- Built internal automation platform that consolidated 4-5 separate tools into one interface, reducing manual work for network operations teams
- Started as Python/Django monolith, later split into services as the system grew and more teams adopted it
- Migrated frontend from Django templates to Angular; built reusable components used by other developers on the team
- Contributed to org-wide migration from Jenkins to GitLab CI/CD, writing pipeline configurations in YAML
- Set up Hugo-based documentation site for spectrum negotiation docs so non-technical stakeholders could maintain them

Computer Science Instructor (Part-time) | Nucamp Coding Bootcamp — *Seattle, WA*

Jan 2018 – May 2018

- Taught full-stack web development (JavaScript, HTML/CSS, Git, REST APIs) to students transitioning into tech careers

Salesforce Engineer | Slalom Consulting — *Seattle, WA*

Nov 2017 – Mar 2018

- Earned three Salesforce certifications (Integration Architect Designer, Administrator, Platform App Builder) and worked on CRM implementations for enterprise clients

Co-Founder & Technical Lead | OLI Fitness — *Seattle, WA*

Jan 2016 – Mar 2018

- Co-founded fitness startup focused on preventing injuries during weightlifting. Built real-time joint tracking using Microsoft Kinect SDK (C#) to analyze squat and deadlift form
- Developed scoring system comparing user joint angles to "good form" parameters defined with a certified personal trainer
- Published extended abstract at ACM CHI 2017. Princeton Tiger Launch regional finalist. UW Business Plan Competition finalist. Selected for Collision Alpha startup program

Software Engineering Intern | Givesafe (now Samaritan) — *Seattle, WA*

Dec 2016 – Apr 2017

- Built features in Ruby on Rails for a platform connecting donors with people experiencing homelessness

PERSONAL PROJECTS

Old Ways Today — *AI Product Search Platform*

2025 – Present

- Full-stack platform helping families find non-toxic household products. Built semantic search using OpenAI embeddings (text-embedding-3-small) with server-side cosine similarity ranking
- Includes blog CMS with markdown editor, admin dashboard, email subscription system with SMTP, and Amazon affiliate integration
- React frontend on Netlify, FastAPI/PostgreSQL backend on Render. Automatic Twitter postings

azoni.ai — *Portfolio with AI Assistant*

2025 – Present

- Personal portfolio site featuring an AI chatbot trained on my background via custom system prompts
- Recruiters can paste job descriptions and the chatbot explains my fit.
- React with modern patterns (Context, Hooks, Suspense, Error Boundaries). Secure API handling via Netlify Functions

Rowing Tracker — *AI Fitness App*

2025 – Present

- App that extracts workout data from rowing machine photos using Claude's vision API. Users verify extracted numbers before saving
- Includes leaderboards, and achievements. Actively maintaining and improving based on user feedback

AI Tools & Apps — *Bots and Prediction Market*

2024 – Present

- DuMarket — Prediction market webapp with order book matching (price-time priority), automated market maker with inventory-aware pricing, and P&L tracking. React, FastAPI, PostgreSQL
- LLM-Powered Bots — Discord and Twitter bots with persistent memory (SQLite), tool integrations (Google Calendar, web search), and agentic decision-making using GPT-4 and Claude APIs

EDUCATION

M.S. Software Engineering — Colorado Technical University

2021

Completed while working full-time at T-Mobile

B.S. Computer Science — University of Washington Tacoma

2017

Graduated with Honors • Teaching Assistant (Java, Python)

RECOGNITION & COMMUNITY

- 1st Place, T-Mobile Big Data Hackathon — Built auto-hashtag tool using OpenCV and geolocation
- ACM CHI 2017 Publication — "OLI Fitness" extended abstract on computer vision for injury prevention
- Organized Global AI Hackathon Seattle 2017 — Coordinated sponsors, judges, and volunteers
- President, Huscii Coding Club (UW Tacoma) — Led weekly workshops, organized hackathon teams
- Mentor, Expedia Coding for Kids — Weekly volunteer teaching elementary students programming basics